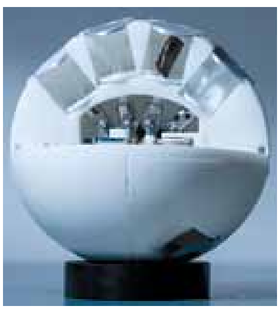
An energy-generation device turns vibrations into power

Researchers at Tohoku University have developed a new energy-generation device by merging piezoelectric composites with carbon fibre-reinforced polymer (CFRP), a lightweight and robust material. The researchers considered using a durable CFRP-based piezoelectric vibration energy harvester (PVEH) to explore an efficient and enduring energy harvesting method. They combined potassium sodium niobate (KNN) nanoparticles with epoxy resin, incorporating them into CFRP to create the device. CFRP served as both an electrode and reinforcement during fabrication. The C-PVEH device surpassed expectations, enduring 100,000 bends while maintaining high performance. It stored electricity efficiently and powered LED lights. It outperformed other KNN-based composites in energy output density. The C-PVEH's introduction will advance self-powered IoT sensors, promoting energy-efficient IoT devices. The high energy output and durability of the device pave the way for exploring composite materials in various applications.

The principle, structural design, and application of carbon fibre-reinforced polymer-enhanced piezoelectric nanocomposite materials (Credit: Tohoku University)

An origami-based haptic device to enhance virtual reality experiences

Researchers at Westlake University, the Westlake Institute for Advanced Study, and other universities in China have developed an innovative haptic device that can enhance the fidelity and realism of virtual experiences. The origami-based device enables active interaction with virtual objects, providing immersive touch experiences. The device synchronises with virtual reality (VR) content, delivering precise tactile feedback aligned with user interactions. It replicates sensations of different object textures and even crushing experiences. The system features a curved origami panel that adjusts stiffness using motors, enabling dynamic modifications in VR/AR environments. This creates immersive, tactile experiences termed 'active mechanical sensation.' Initial tests demonstrated the device's superiority over vibration-based haptic devices in generating realistic touch sensations. The origami-based design has the potential to enhance VR performance.



An origami-based haptic device (Credit: Danyang Zhu, Westlake University)

Holy grail: An electric vehicle technology unlocks six minutes charge time

Nyobolt, a startup based in the UK, asserts a significant advancement in battery technology, potentially enabling EVs to recharge in under ten minutes. The company proclaimed a battery breakthrough called the "holy grail," enabling EVs to charge fully in less than six minutes. They plan to produce this technology by 2024, reducing range anxiety and enabling smaller battery packs. Nyobolt collaborates with Callum design firm to develop an electric sports car. Lightweight carbon-fibre bodywork reduces weight to 1,000kg. It features a 35kWh battery pack and claims 250km range with a six-minute charge, adding 2,500km per hour of charging. Production plans for the sports car remain undisclosed. The startup claims readiness for the rapid scale-up of battery technology. Extensive testing, including 2,000+ fast-charge cycles, showed no notable performance decline.

Can robots learn from videos?

Researchers at Carnegie Mellon University have enabled robots to learn household chores by watching home videos of people performing everyday tasks. The researchers have enhanced home robot utility, enabling cooking, cleaning, and

A team from Carnegie Mellon University's Robotics Institute used affordances to teach robots how to interact with objects (Credit: Carnegie Mellon University)